

Artificial Intelligence Information Retrieval

Project #1 Keyword-based Full-Text Matching

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1. 系統目的

輸入 PMID 完成 PubMed Abstract 內正文字元、字、句統計與檢索。

2. 統計範圍定義

以 PubMed Abstract 可見範圍為統計語料，包含區段標籤；Keywords 不納入統計。

Andrological effects of SARS-Cov-2 infection: a systematic review and meta-analysis

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Affiliations + expand

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Abstract

Purpose: The short- and long-term andrological effects of coronavirus disease 2019 (COVID-19) have not been clarified. Our aim is to evaluate the available evidence regarding possible andrological consequences of COVID-19 either on seminal or hormonal parameters. The safety of the COVID-19 vaccines in terms of sperm quality was also investigated.

Methods: All prospective and retrospective observational studies reporting information on severe acute respiratory syndrome coronavirus 2 (SARS-Cov-2) mRNA semen and male genitalia tract detection (n = 19), as well as those reporting data on semen analysis (n = 5) and hormonal parameters (n = 11) in infected/recovered patients without any arbitrary restriction were included.

Results: Out of 204 retrieved articles, 35 were considered, including 2092 patients and 1138 controls with a mean age of 44.1 ± 12.6 years, and mean follow-up 24.3 ± 18.9 days. SARS-CoV-2 mRNA can be localized in male genitalia tracts during the acute phase of the disease. COVID-19 can result in short-term impaired sperm and T production. Available data cannot clarify long-term andrological effects. Low T observed in the acute phase of the disease is associated with an increased risk of being admitted to the Intensive Care Unit or death. The two available studies showed that the use of mRNA COVID-19 vaccines does not affect sperm quality.

Conclusions: The results of our analysis clearly suggest that each patient recovering from COVID-19 should be monitored to rule out sperm and T abnormalities. The specific contribution of reduced T levels during the acute phase of the infection needs to be better clarified.

Keywords: COVID-19; Hypogonadism; SARS-CoV-2; Sperm; Testosterone; Vaccination.

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PubMed Disclaimer

3. 統計方式

Characters

含空白：將連續空白、換行與 Tab 統一為單一空格，移除首尾空白後，以 `len(text)` 計算字元數。

```
# Characters with spaces
character_count = len(text)
```

不含空白：在上述文字中移除所有空白字元後再計數。
兩者皆保留標點與數字。

```
# Characters without spaces
character_count_without_spaces = len(
    re.sub(r"\s+", "", text)
)
```

Words

採 whitespace segmentation，以 `len(text.split())` 計數。
不拆開詞內符號：如 cancer-dementia、16%/9% 各算 1 詞。
測試後與 MS Word 計數較一致，因此統計採 Whitespace。

```
# Words
words = text.split()
word_count = len(words)
```

Sentences

採 pySBD 進行英文句界判定，處理縮寫、小數與句末標點；各摘要區段獨立分句後加總。
依標籤辨識的註冊編號等非敘述性區段不計入句數，但仍計入字元與詞數。

```
SENTENCE_SEGMENTER = pysbd.Segmenter(
    language="en",
    clean=False
)

sentences = SENTENCE_SEGMENTER.segment(text)
sentence_count = len(sentences)
```

pySBD 簡介

一個 Python 套件，全名是 Python Sentence Boundary Disambiguation。它是一套 rule-based 的句子邊界判定工具，不是機器學習模型。

它的核心來源是 Ruby 的 Pragmatic Segmenter，後來移植到 Python，並加入額外改善。

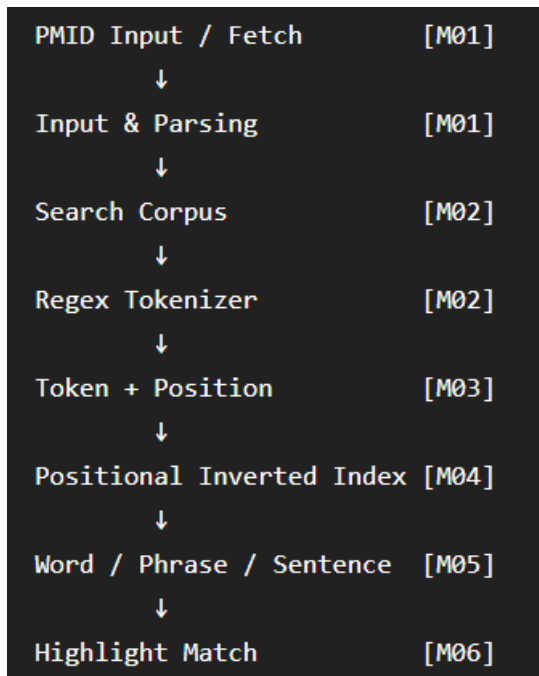
2020 年有對應論文發表在 EMNLP 的 NLP Open Source Software Workshop。

4. 搜尋方法

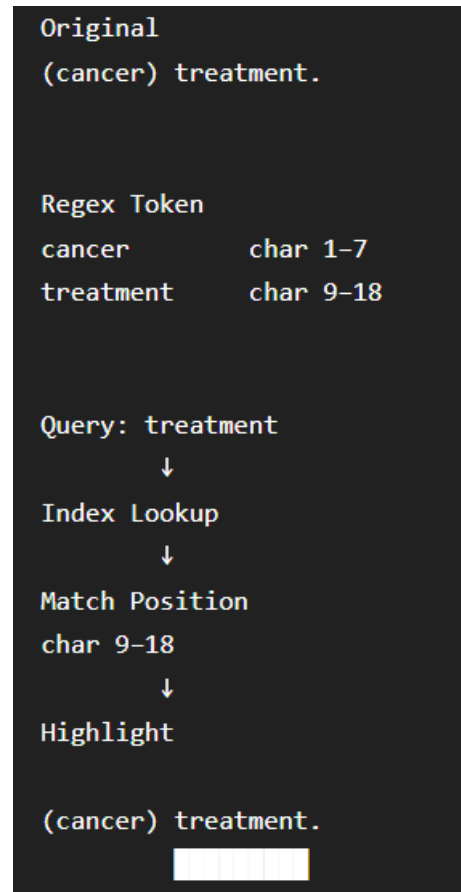
搜尋另建 Search Corpus，採 Regex tokenizer 切詞；後續記錄 token 位置並建立位置式倒排索引。

單詞使用索引比對；片語與句子在句內依相鄰、鄰近距離或詞序條件匹配，並以字元位置反白命中。

檢索流程概覽 ↓



Regex Token 與 Position 範例 ↓



5. 系統架構

系統以 Python 實作，分為 6 個功能模組（M01→M06）

